

BUSINESS DECISION COMPARISON REPORT: LUXURY HIGH-RISE FOUNDATION & PODIUM PACKAGE

Prepared by Proposal Compare | General Construction Vertical

BUSINESS DECISION

BlueRock Institutional Partners LP (Blackstone JV), via BrightPath Property Management, is managing construction of Lumina Hotel & Residences – 42-story luxury mixed-use tower (Four Seasons operator, 198 hotel keys, 140 condos, 18k retail) at 1800 West Loop South, Houston. The owner is evaluating three bids for a modest \$100k-range sub-scope: Phase 1A Foundation & Podium Transfer – 12 drilled piers, grade beams, post-tensioned podium slab Level 2, waterproofing, temporary dewatering. This is a critical path sub-package supporting tower crane erection; failure impacts \$140M project. Buyer priorities (per intake): institutional-grade reporting for PE investors, schedule certainty to avoid tower crane standby (\$15k/day), structural longevity (100-year PT life in Gulf Coast clay), risk transfer for unforeseen obstructions and dewatering, compliance with 2021 IBC + ACI 318 + Texas geotech, and cost discipline for this slice while maintaining luxury quality. No vendor preference. This report is objective feature-by-feature analysis, no scoring, no recommendation.

PROPOSAL COMPARISON

FEATURE: Structural Engineering & Concrete Mix Design

Proposal A – Atlas (\$98,750): 36-inch belled to 60-inch @ 38-42 ft, 5000 psi piers / 6000 psi podium, SCC mix 15% fly ash, 0.42 w/c, Grade 60 rebar #6 @12-inch, 12 bars pier, 0.5-inch 270K unbonded PT 33 tendons (?) 1.2 tons, 2-stage stressing. Blueskin WP200 membrane.

Proposal B – Meridian (\$112,400 base after credits): Same belled geometry, upgraded to 5500 psi piers with Sika CNI corrosion inhibitor / 6500 psi podium + silica fume + macro+micro fibers, Grade 80 rebar #6 @10-inch, 14 bars pier, 0.6-inch DSI encapsulated PT higher capacity, 3-stage with lift-off test, 12 ConcreteSense IoT maturity sensors, Carlisle MiraDRI 80-mil + duplex sump with BMS, 2 confirmation borings included.

Proposal C – Sterling Forge (\$87,200): Value-engineered straight shaft (no bell) 36-inch to 48 ft deep (deeper to compensate), 4500 psi piers / 6000 psi podium, non-SCC, 0.45 w/c, Grade 60, #6 @12-inch, 10 bars pier, 0.5-inch PT 28 tendons (reduced), 11-inch slab vs 12-inch, precast U-shell grade beams + infill, Voltex bentonite waterproofing + single drain.

SIGNIFICANCE: Meridian offers highest durability (corrosion inhibitor, fibers, Grade 80, higher psi) relevant for 100-year luxury asset and Houston clay corrosion. Atlas is middle-ground with SCC for lower heat of hydration (reduces cracking risk). Sterling's straight-shaft alternative avoids bell risk in caving soils but requires deeper embedment (48 ft vs 40 ft) – if bearing stratum deeper than anticipated, extra \$95/LF change order risk. Reduced slab thickness (11-inch vs 12-inch) and fewer tendons saves cost but changes stiffness for transfer loads – requires SER approval. Buyer should verify SER acceptance of straight shaft vs belled and 11-inch vs 12-inch for retail loads below.

FEATURE: Means & Methods, Schedule & Technology

Proposal A: Bauer BG30 rig + belling tool, 50T crane, 36m boom pump, single 280 CY monolithic night pour, 28 days, 5 weather days, LD \$500/day capped 10%, early bonus \$250/day. Peri Skydeck formwork. No IoT, no drone.

Proposal B: Soilmec SR-75 + polymer slurry, 42m pump, Peri Duo aluminum, ConcreteSense IoT maturity allows early stressing (75% f'c in 3 days vs 7 days), 2 pours (300 CY total), 21 days (7 days faster), 10 weather days, automated dewatering telemetry 4 wells 100 gpm + frac tank, drone daily, 4D BIM P6 + Procore EV, 4 meetings, GPR scan for tendon verification.

Proposal C: Texoma 650 smaller rig, straight shaft (no belling tool), 25T crane, 32m pump, 11-inch slab 260 CY (8% less concrete), 32 days (longest), 7 weather days, no LD, 5% retention to 60 days, precast shells speed grade beam 2 days but deeper piers slower (1.5/day vs 2/day), dewatering 2 wells passive 40 gpm no telemetry.

SIGNIFICANCE: Schedule certainty critical for institutional investor – Meridian 21-day schedule saves ~7 days vs Atlas and 11 days vs Sterling, potentially saving tower crane standby \$105k+ if on critical path, justifying premium. Atlas night pour controls

heat but single monolithic pour risk (cold joint if truck delayed). Sterling precast shells reduce formwork labor but deeper straight piers add drilling time and risk of caving without slurry. IoT maturity and drone in Meridian provide data for investor reporting – valued by Blackstone JV.

FEATURE: Quality Assurance, Testing & Compliance

Proposal A: Terracon testing slump/air/temp every truck, 6 cylinders/50 CY, 100% PIT, PT elongation, 1 geotech visit, LOD300 BIM, 2 coordination meetings, no CSL, no GPR, no peer review beyond \$1,500 SER allowance.

Proposal B: Terracon + in-house, SCC flow tests (slump flow/J-ring/L-box), air/temp per truck, 8 cylinders/50 CY, maturity calibration, 100% PIT + 2 CSL included (higher reliability), GPR tendon verification, 2 extra borings, LOD400 + Tekla rebar, 4D P6, Walter P Moore peer review \$2,500 included, LEED EPDs, QR codes.

Proposal C: Terracon per 2 trucks (not every truck), 4 cylinders/100 CY, PIT 50% only (6 of 12), no CSL, no GPR, no maturity, 2D CAD only (no BIM), 1 meeting, no peer review (optional \$2,200), no LEED.

SIGNIFICANCE: Testing thoroughness affects risk transfer – Meridian 100% PIT + CSL + GPR provides highest confidence for luxury high-rise foundation (institutional lenders often require CSL). Atlas 100% PIT is industry standard. Sterling 50% PIT reduces cost \$2,500 but increases risk of undetected defect in expansive clay. BIM level matters for MEP penetration coordination in podium (retail plumbing) – LOD400 in Meridian reduces change orders; 2D CAD in Sterling increases risk of missed sleeves. LEED EPDs in Meridian support possible LEED Gold for hotel.

FEATURE: Waterproofing, Dewatering & Long-Term Durability

Proposal A: Blueskin WP200 self-adhesive 6200 sqft + 6mm HDPE + dimple drain + 4-inch perf to sump, temp dewatering 2 wells 20 gpm each 3 weeks, permit included.

Proposal B: Premium Carlisle MiraDRI 860/861 80-mil composite + 1/2-inch HDPE + 1-inch drain mat 10 gpm/ft + dual redundant subdrain + duplex Zoeller sump with BMS alarm, temp dewatering 4 wells automated telemetry 100 gpm total + frac tank 4 weeks, turbidity <50 NTU, duplex pump package.

Proposal C: VE Cetco Voltex DS bentonite geotextile self-healing + 6mm protection + standard dimple board, single 4-inch perf, single sump (no duplex), temp dewatering 2 wells passive 40 gpm no telemetry.

SIGNIFICANCE: Waterproofing is critical for luxury retail below podium (high-end finishes). Carlisle 80-mil with redundant drain and duplex pump (Meridian) offers highest resilience vs single drain/single pump in Sterling. Bentonite (Sterling) is self-healing but can be compromised if not hydrated properly. Dewatering – Meridian 4-well automated system reduces risk of base heave in high water table (-12 ft); passive 2-well in Sterling may be insufficient if water table rises after rain, causing delay. Buyer should verify if permanent sump system (beyond temp) is in other packages – none include permanent beyond Meridian duplex option.

FEATURE: Warranty, Insurance, Risk Transfer & Institutional Reporting

Proposal A: 2-year workmanship, 1-year concrete crack <0.02 inch, 10-year PT tendon supplier, no waterproofing extended. GL \$2M/\$4M Umbrella \$5M, Workers Comp, Professional \$1M for PT. \$500/day LD capped 10%, early bonus. 10% retainage. Performance & Payment bond 2.5% extra. No obstruction allowance.

Proposal B: 4-year workmanship (double), 2-year concrete crack >0.03 inch inject, 15-year PT encapsulated, 5-year Carlisle waterproofing, 6-year if \$1,200/yr inspection. GL \$5M/\$10M Umbrella \$10M Professional \$2M Pollution \$2M Builders Risk \$100k included. \$15k obstruction allowance, dewatering failure risk borne 48hrs backup included, 10 weather days, OCIP/CCIP compatible. No LD but risk register + P6 EV + daily Procure + weekly lender call (2 hrs) included – institutional reporting for PE. Bond 1.8%.

Proposal C: 1-year workmanship (minimum), 1-year concrete, 10-year PT supplier, 2-year Voltex. GL \$1M/\$2M Umbrella \$2M no pollution/builders risk/professional. No LD, 5% retention to 60 days, 10% otherwise. No obstruction allowance – T&M; \$175/hr extra, straight shaft deeper than 48 ft \$95/LF extra change order risk. Bond 3.0% (higher due to smaller balance sheet). No EV reporting, Procure daily only, no P6.

SIGNIFICANCE: For institutional PE owners, risk transfer and reporting matter more than lowest bid. Meridian 4-year warranty + pollution insurance + \$15k obstruction allowance + duplex dewatering backup provides strongest risk transfer. Atlas middle. Sterling lowest insurance limits and no pollution coverage – potential gap if dewatering discharge violates permit. No LD in Sterling may reduce schedule incentive. Bond rate inverse to size – Sterling 3.0% vs Meridian 1.8% indicates financial strength

difference – relevant for \$140M project continuity.

PRICE AND SCOPE ANALYSIS

Proposal A – Atlas: \$98,750 net after 2% early pay discount (-\$3,000) from \$101,750 subtotal. Breakdown: piers \$38,500 (38.9%), grade beams \$12,800, podium PT \$32,500, waterproofing \$8,200, dewatering \$3,200, BIM/engineering \$2,400, mob \$4,150. Includes 100% PIT, SCC mix, Blueskin, 28 days, 2-yr warranty. Excludes soil export beyond allowance, elevator pit, mass excavation beyond -10 ft, permanent dewatering. Price lock 60 days then ACI index escalation.

Proposal B – Meridian: \$112,400 after credits: subtotal \$122,400 minus \$4k power credit minus \$6k early award credit. Piers \$44,200 (+\$5,700 vs Atlas due to corrosion inhibitor + CSL + extra borings), grade beams \$14,500 (+\$1,700 Grade80 + stainless chairs), podium \$36,800 (+\$4,300 fibers + 6500 psi + maturity + 0.6-inch strand), waterproofing \$11,200 (+\$3,000 premium Carlisle + duplex), dewatering \$5,500 (+\$2,300 telemetry + frac tank), BIM 4D \$4,800 (+\$2,400 vs Atlas), mob/tech \$5,400 (+\$1,250 drone/GPR). Options: Osterberg load test \$12k, cathodic anodes \$3,800. Includes 21 days (7 days faster), 4-yr warranty, institutional reporting. Price valid 45 days then ENR CCI escalation.

Proposal C – Sterling Forge: \$87,200 firm 90 days (best price lock). Subtotal \$79,800 + \$2,400 contingency + \$10k OH&P; 12%. Piers \$31,500 (-\$7k vs Atlas due to 4500 psi, fewer bars, 50% PIT, straight shaft no bell), grade beams precast \$10,200 (-\$2,600), podium \$26,800 (-\$5,700 11-inch + fewer tendons), waterproofing \$5,200 (-\$3k Voltex), dewatering \$2,100 (-\$1,100 passive), engineering \$900, mob \$3,100. Savings from VE: no bell tool \$3,500, precast shells \$2,800, 8% less concrete \$1,200, fewer tendons \$1,800, cheaper waterproofing \$3k, reduced testing \$2,500, 2D vs BIM \$1,500, lower insurance. Options: upgrade to 100% PIT+2 CSL \$2,800, duplex sump \$1,900.

Price Difference Drivers: Range \$87,200 to \$112,400 = \$25,200 spread (28.9% variance). Lowest (Sterling) achieves savings via straight shaft (no bell), precast grade beams, thinner slab (11-inch), fewer PT tendons, VE waterproofing, reduced testing (50% PIT), 2D CAD, smaller equipment, lower insurance limits. Middle (Atlas) is baseline industry standard with SCC and 100% PIT. Highest (Meridian) premium justified by: corrosion inhibitor + fibers + Grade80 + higher psi, 0.6-inch encapsulated PT + maturity sensors, premium Carlisle waterproofing with redundant duplex, 4-well automated dewatering, CSL + GPR, LOD400 + 4D P6 + peer review, 4-year warranty + pollution + builders risk, 7-day schedule acceleration (value ~\$105k in tower crane savings), and institutional reporting (daily Procore + lender calls). Recurring: No recurring fees in any, but Meridian extended inspection \$1,200/yr optional. Buyer should normalize for testing (add \$2,800 to Sterling for 100% PIT+CSL = \$90k), for dewatering backup risk, and for potential change order if straight shaft needs deeper than 48 ft.

IMPORTANT UNCERTAINTIES & OMISSIONS

1. Soil export and contamination: All exclude extensive soil haul-off; if borings show contaminated fill (Houston historic), disposal cost \$85-\$120/CY not included – who bears?
2. Straight shaft vs belled acceptance: Sterling 48 ft straight shaft alternative requires SER approval; geotech report allows it but SER has not approved – risk of redesign to belled if SER rejects.
3. Bearing stratum depth: If competent stratum deeper than 40-42 ft (Atlas/Meridian) or 48 ft (Sterling), extra pier length at \$95-\$125/LF change order – allowance unclear.
4. MEP penetrations: Podium slab has retail plumbing sleeves – BIM LOD300/400 in Atlas/Meridian includes coordination; Sterling 2D only – risk of missed sleeves causing core drilling \$400 each.
5. Waterproofing warranty transfer: Carlisle 5-year (Meridian) vs Voltex 2-year (Sterling) – for luxury retail below, 2-year may be insufficient for institutional lender.
6. Dewatering permit turbidity: City of Houston requires <50 NTU – Meridian includes monitoring, others basic – risk of stop-work if turbidity exceeded.
7. Tower crane foundation: Podium transfer slab must support crane – none explicitly include crane load analysis – confirm if crane sits on podium or separate.
8. LD and schedule risk: Sterling has no LD and 32 days – if delay impacts \$140M project, owner bears tower crane standby cost \$15k/day – insurance gap.

QUESTIONS FOR THE VENDORS

To Atlas (Proposal A):

1. Can you include 2 CSL tests like Meridian for same \$38,500 pier cost, or price add? Provide SCC mix EPD for LEED and confirm night pour cold joint mitigation plan (backup pump).
2. What is cost to upgrade waterproofing to Carlisle 80-mil + duplex sump with BMS like Meridian? Also confirm dewatering backup pump on-site if failure.
3. Can you extend workmanship warranty to 4 years to match Meridian and include weekly lender call participation for BlueRock

reporting?

To Meridian Elite (Proposal B):

1. Your 21-day schedule is aggressive – can you provide P6 showing float and guarantee tower crane ready date? What LD or credit if you miss 21 days, given you currently propose no LD?
2. Confirm \$15k obstruction allowance covers old foundations – Houston CBD often has obstructions deeper than 10 ft. Is \$95/LF pier extra beyond 42 ft included or separate?
3. Can you hold price 60 days firm without ENR escalation to match Atlas/Sterling 60/90 day lock, given institutional budgeting needs?

To Sterling Forge (Proposal C):

1. Straight shaft 48 ft vs belled 40 ft – can you provide SER pre-approval letter for straight shaft alternative and confirm if extra depth beyond 48 ft is at your risk or T&M? Also upgrade to 100% PIT + 2 CSL for \$2,800 – can you include at \$87,200?
2. Waterproofing Voltex 2-year vs Carlisle 5-year – for luxury retail with marble flooring below, can you upgrade to Carlisle + duplex sump with BMS at what cost, and extend waterproofing warranty to 5 years?
3. Insurance limits \$1M/\$2M Umbrella \$2M are below lender requirement (\$2M/\$4M min) – can you increase to \$2M/\$4M + Umbrella \$5M and add pollution coverage for dewatering without increasing bid, or at what add?

To All:

4. Provide 3 references for similar luxury hotel podium (Four Seasons/Hyatt/Ritz) with owner contact and confirm if any had dewatering challenges in Gulf Coast clay.
5. Confirm if tower crane (300-ton) will sit on this podium slab – if so, provide analysis and any strengthening required, or state if separate crane mat is by others.

BOTTOM LINE

Three bids for same Phase 1A foundation & podium scope range \$87,200 to \$112,400. Proposal A Atlas (\$98,750) offers industry-standard belled piers, SCC mix, Blueskin waterproofing, 28-day schedule, 100% PIT, 2-year warranty – balanced middle cost with night pour heat control and 2% early pay discount. Proposal B Meridian (\$112,400) offers premium institutional-grade approach: higher psi + corrosion inhibitor + fibers + Grade80 rebar, 0.6-inch encapsulated PT with IoT maturity, Carlisle 80-mil + duplex BMS sump, 4-well automated dewatering, 100% PIT + 2 CSL + GPR, LOD400 + 4D P6 + peer review, 21-day accelerated schedule (7-11 days faster), 4-year warranty + pollution + builders risk, and daily Procore + lender call reporting – premium justified by schedule savings (\$105k+ crane standby avoidance) and risk transfer for PE owners, but price valid only 45 days with ENR escalation. Proposal C Sterling Forge (\$87,200) offers value-engineered lowest cost via straight shaft no bell (deeper 48 ft), precast grade beam shells, 11-inch slab vs 12-inch, 28 vs 33-36 tendons, Voltex bentonite + single drain, 50% PIT, 2D CAD – saves ~\$18k vs Atlas but introduces SER approval risk for straight shaft, reduced testing, thinner slab, lower insurance limits (\$1M/\$2M vs \$2M/\$4M), no pollution coverage, no LD, and 32-day schedule (longest). Tradeoffs center on durability/longevity vs cost discipline, schedule certainty vs savings, and institutional reporting vs basic daily reports. Unresolved: soil export contamination, bearing depth extra length, MEP sleeve coordination, waterproofing warranty adequacy for luxury retail, and tower crane load on podium. No vendor recommendation made – decision should align with BlueRock risk tolerance, lender insurance requirements, and importance of 7-11 day acceleration to \$140M project.

Disclaimer: This report is decision support only and does not constitute engineering, legal, financial, or construction advice. Verify all structural, geotech, and code claims with SER and licensed professionals.